

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination January 2018

PHPT001 Novel Drug Delivery Systems-I

Date: 05/01/2018, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** A) Write note on techniques used for permeation enhancement of drug in transdermal drug delivery systems. 05
B) Explain the evaluation parameters for fast dissolving films. 05
- Q.2** A) Describe the model dependent methods to investigate the kinetics of drug release from controlled release formulations. 05
B) Discuss the factors affecting the bioadhesion of mucoadhesive drug delivery systems. 05

OR

- Q.2** A) Discuss in detail the rationale behind development of gastroretentive drug delivery systems. 05
B) Discuss the use of microemulsion and/or nanoemulsions as a drug delivery system. 05
- Q.3** Answer the following (Any three) 15
A) Write note on sensor based implants.
B) Write note on use of polymers for the development of hydrogels.
C) Discuss the components required for the development of transdermal patch.
D) Write note on use of monoclonal antibodies for drug targeting.

SECTION-II

- Q.4** A) Discuss the use of superdisintegrants in formulation of fast disintegrating tablets. 05
B) Write note on characterization of liposomes. 05
- Q.5** A) Explain in detail the formulation of osmotically controlled drug delivery system. 05
B) Describe the methods of crosslinking for the formulation of hydrogels. 05

OR

- Q.5** A) Describe the methods of preparation for polymeric nanoparticles. 05
B) Discuss basic components required for the development of buccal mucoadhesive drug delivery systems. 05
- Q.6** Answer the following (Any three) 15
A) Write note on ophthalmic inserts.
B) Explain the fundamentals of drug permeation through the skin.
C) Describe in brief the procedure for formulation and optimization of nanoemulsions formulation.
D) Discuss the applications of hydrogel in drug delivery.